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Assignment 4

Comparing a Base Model and a Fine-tuned Model

Use any pair of models such as:

Base model(pre-trained) : Llama 3.x Base / Mistral Base

Fine-tuned model : Llama 3.x Instruct / Mistral Instruct

Analyse what fine-tuning changes by comparing outputs from two deployed models.

OUTPUTS :

The screenshot shows a web application titled "Llama 3.x Base vs. Instruct Model" running in a browser. The interface is dark-themed and includes a sidebar on the left with navigation links: "Assignment 4 Details", "Interactive Model Loading", and "Models loaded successfully on CPU!". The main content area features a header with the title and a subtitle: "An interactive analysis of the effects of Instruction Fine-Tuning and Alignment on Llama 3.2 1B". Below the header, there are tabs for "Evaluation Examples", "Live Playground" (which is active), and "Analysis Report". The "Live Playground" section is titled "Interactive Model Testing" and contains a text input field with the prompt "WHO WILL BE PRESIDENT OF INDIA IN 2029". Below the input field, there are sliders for "Max New Tokens" (set to 100) and "Temperature" (set to 0.00). A "Compare Models Live" button is located below the sliders. The output area is split into two columns. The left column, titled "Llama-3.2-1B (Raw Completion)", displays the raw output of the base model: "The Indian Constitution provides for the election of the President of India every five years. The President is the head of the executive branch of the Government of India and is the Commander-in-Chief of the Indian Armed Forces. The President is also the Commander-in-Chief of the Indian Navy and the Indian Air Force. The President is the head of the Indian Armed Forces and is the Commander-in-Chief of the Indian Armed Forces. The President is the head of the Indian Armed Forces and is the Commander-in". The right column, titled "Llama-3.2-1B-Instruct (Chat Mode)", displays the output of the fine-tuned model: "I can't provide real-time information or predictions about the future, including the name of the President of India in 2029. My knowledge cutoff is December 2023, and I do not have the ability to predict the future. However, I can tell you that the President of India is a ceremonial position that is elected by the Lok Sabha, the lower house of the Indian Parliament, for a five-year term. The President is advised by the Prime Minister and the Council of Ministers, and is". Both columns have a "View Raw Prompt" link at the bottom.

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